_eda_price_2025

March 31, 2026

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import subprocess
import os
from scipy.stats import chi2_contingency
from datetime import timedelta

from etl.customer import aggregate_customer
from etl.subsidy import report_customer_conversion_by_subsidy_issue_status,␣
↪report_customer_metrics_by_subsidy_issue_status
from etl.subsidy import visualize_conversion_by_subsidy_issues,␣
↪report_customer_subsidy_issues_by_product
from etl.subsidy import visualize_customer_subsidy_issues_by_product,␣
↪visualize_policy_suspension_impact
from etl.price import visualize_conversion_by_price_sweet_spots,␣
↪visualize_conversion_by_price_sweet_spots_variant
from etl.price import report_product_conversion_by_price_quartile,␣
↪visualize_conversion_by_equipment_category_price
from etl.price import visualise_conversion_by_brand_price,␣
↪show_two_panel_conversion_by_brand_price
from etl.price import visualize_conversion_by_equipment_type_brand_price,␣
↪show_conversion_by_price_dashboard,␣
↪show_product_type_price_conversion_heatmap
from etl.season import visualize_conversion_by_season,␣
↪visualize_regional_seasonality
from etl.period import visualize_conversion_by_year,␣
↪report_best_and_worst_months, show_conversion_by_price_over_time,␣
↪overlay_all_years_on_one_plot
from etl.period import visualize_conversion_by_prices_sweet_spot_shift,␣
↪visualize_decision_times, show_decision_time_dashboard
from etl.product import visualise_heat_pump_performance,␣
↪show_product_type_by_brand_heatmap, show_product_type_by_agency_heatmap,␣
↪show_brand_by_product_type_heatmap
```

```python
import warnings
warnings.filterwarnings('ignore')

df_clean = pd.read_csv('cleaned_quote_data.csv')
df_clean['dt_creation_devis'] = pd.to_datetime(df_clean['dt_creation_devis'])
print(f"Loaded data: {len(df_clean):,} quotes from {df_clean['numero_compte'].
 ↪nunique():,} customers")
```

```
Loaded data: 37,008 quotes from 25,716 customers
```

```python
#customers = aggregate_customer(df_clean)
```

```python
all_customers = pd.read_csv('customer_master_data.csv')
```

```python
# Method 1: Using year column
customers_2025 = all_customers[all_customers['year'] == 2025]

# Method 2: Using period column
customers_2025 = all_customers[all_customers['period'] == '2025']

# Method 3: Using date range
customers_2025 = all_customers[
    (all_customers['first_quote_date'] >= '2025-01-01') &
    (all_customers['first_quote_date'] <= '2025-12-31')
]

print(f"2025 customers: {len(customers_2025):,}")
```

```
2025 customers: 9,986
```

```python
customers = customers_2025
```

```python
conv_no_issue, conv_with_issue =␣
 ↪report_customer_conversion_by_subsidy_issue_status(customers)
report_customer_metrics_by_subsidy_issue_status(customers)
visualize_conversion_by_subsidy_issues(customers, conv_no_issue,␣
 ↪conv_with_issue)
visualize_policy_suspension_impact(customers, price_var='max_out_of_pocket')

top_products, no_issue_pct, issue_pct =␣
 ↪report_customer_subsidy_issues_by_product(customers)
visualize_customer_subsidy_issues_by_product(customers, top_products,␣
 ↪no_issue_pct, issue_pct)

visualize_conversion_by_price_sweet_spots(customers,␣
 ↪price_var='max_out_of_pocket')
```

```
visualize_conversion_by_price_sweet_spots_variant(customers,␣
 ↪price_var='max_out_of_pocket')
report_product_conversion_by_price_quartile(customers)
visualize_conversion_by_equipment_category_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualise_conversion_by_brand_price(customers, price_var='max_out_of_pocket')
show_two_panel_conversion_by_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_equipment_type_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')

visualize_conversion_by_season(customers, price_var='max_out_of_pocket')
visualize_regional_seasonality(customers, price_var='max_out_of_pocket')

visualize_conversion_by_year(customers, price_var='max_out_of_pocket')
report_best_and_worst_months(customers, price_var='max_out_of_pocket')
show_conversion_by_price_over_time(customers, price_var='max_out_of_pocket')
overlay_all_years_on_one_plot(customers, price_var='max_out_of_pocket')
visualize_conversion_by_prices_sweet_spot_shift(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_decision_times(customers, price_var='max_out_of_pocket')
show_product_type_price_conversion_heatmap(customers,␣
 ↪price_var='max_out_of_pocket')
show_decision_time_dashboard(customers, price_var='max_out_of_pocket')

visualise_heat_pump_performance(customers, price_var='max_out_of_pocket')

show_conversion_by_price_dashboard(customers, price_var='max_out_of_pocket')

show_product_type_by_brand_heatmap(customers)
show_product_type_by_agency_heatmap(customers)
show_brand_by_product_type_heatmap(customers)
```

```
Customer conversion by subsidy issue status:

                   conversion_rate  customer_count  converted_count
had_subsidy_issue
False                     0.356779            7464           2663.0
True                      0.450436            2522           1136.0

Chi-square p-value: 6.714639e-17
Statistically significant: YES

Conversion without issues: 35.68%
Conversion with issues: 45.04%
Difference: 9.4 percentage points
```

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Customer-level finding: +9.4 points

================================================================================
Customer characteristics by subsidy issue status
================================================================================

Metric comparison:

total_quotes:
  No issues: 1.40
  With issues: 1.59
  Ratio: 1.14x

avg_quote_amount:
  No issues: 5835.41
  With issues: 9668.17
  Ratio: 1.66x

max_quote_amount:
  No issues: 6240.72
  With issues: 10588.18
  Ratio: 1.70x

customer_duration_days:
  No issues: 9.13
  With issues: 13.12
  Ratio: 1.44x

price_range:
  No issues: 892.37
  With issues: 1733.21
  Ratio: 1.94x

multiple_quotes:
  No issues: 0.29
  With issues: 0.38
  Ratio: 1.32x

Quote pattern distribution by subsidy issue:
had_subsidy_issue  False  True
quote_pattern
Single quote       0.710  0.616
Three+ quotes      0.070  0.122
Two quotes         0.221  0.262

Decision speed by subsidy issue:
had_subsidy_issue   False  True
decision_speed
```

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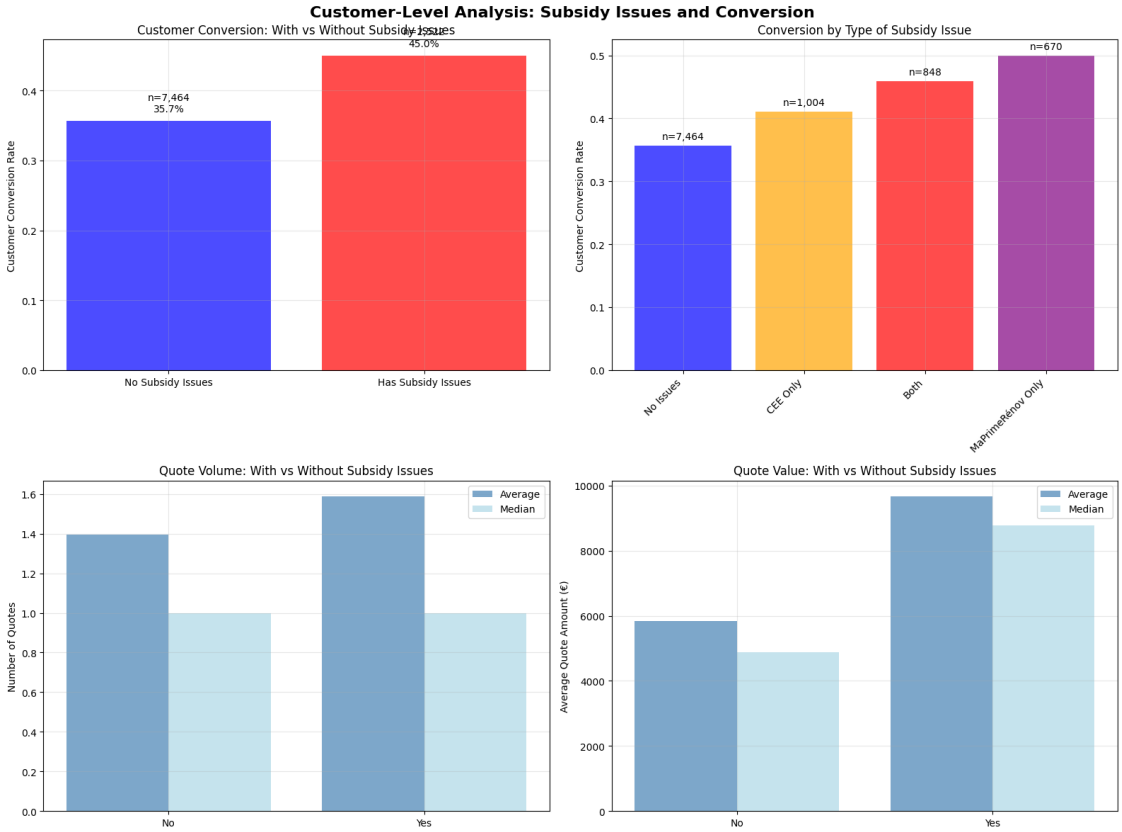
Long (>30 days)     0.058  0.097
Medium (8-30 days)  0.036  0.047
Quick (2-7 days)    0.043  0.073
Same day decision   0.863  0.782

Product family by subsidy issue status:
had_subsidy_issue                 False  True
main_product_family
Accessoire de pose                0.004  0.001
Appareil hybride                  0.003  0.036
Autres                            0.010  0.004
Chaudière                         0.369  0.054
Climatisation                     0.296  0.106
ECS : Chauffe-eau ou adoucisseur  0.066  0.021
Emetteur de chauffage  ou chappe  0.011  0.007
Photovoltaïque                    0.029  0.014
Plomberie Sanitaire               0.001  0.000
Pompe à chaleur                   0.049  0.397
Poêle                             0.150  0.361
Produit VMC                       0.004  0.000
Unknown                           0.008  0.000

Conversion by subsidy issue type:
                        mean  count
subsidy_issue_type
MaPrimeRénov Only   0.500000    670
Both                0.458726    848
CEE Only            0.410359   1004
No Issues           0.356779   7464
```

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================================================================================
Policy Suspension Impact
================================================================================

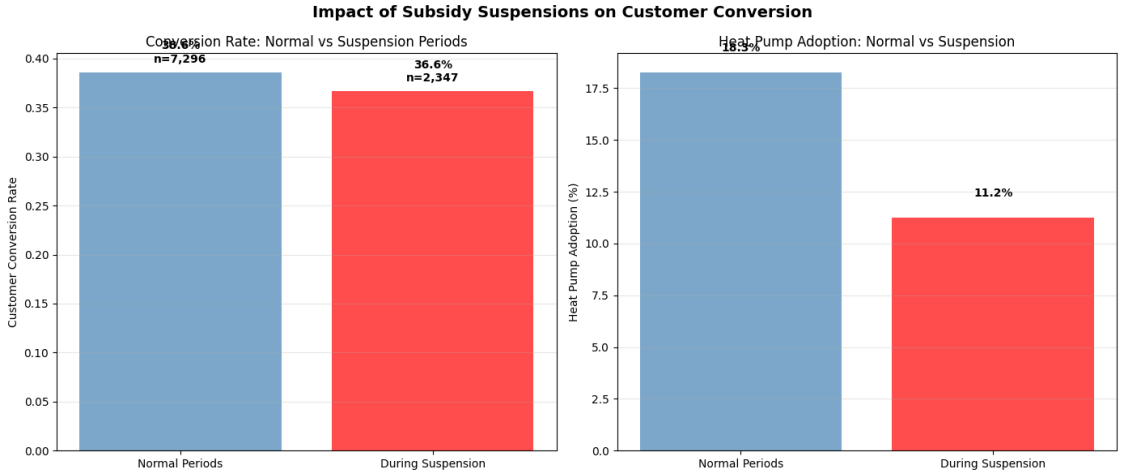
Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
Unique values in during_suspension: [False  True]

Conversion during subsidy suspensions:
                      mean  count
Normal Periods     0.386102   7296
During Suspension  0.366425   2347

Suspension impact p-value: 0.0924
Statistically significant: NO

Product mix during suspensions:
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
during_suspension
```

```
False                    19.9    31.4       15.7   11.2   21.8
True                     44.1    24.2        8.9    7.4   15.4
```



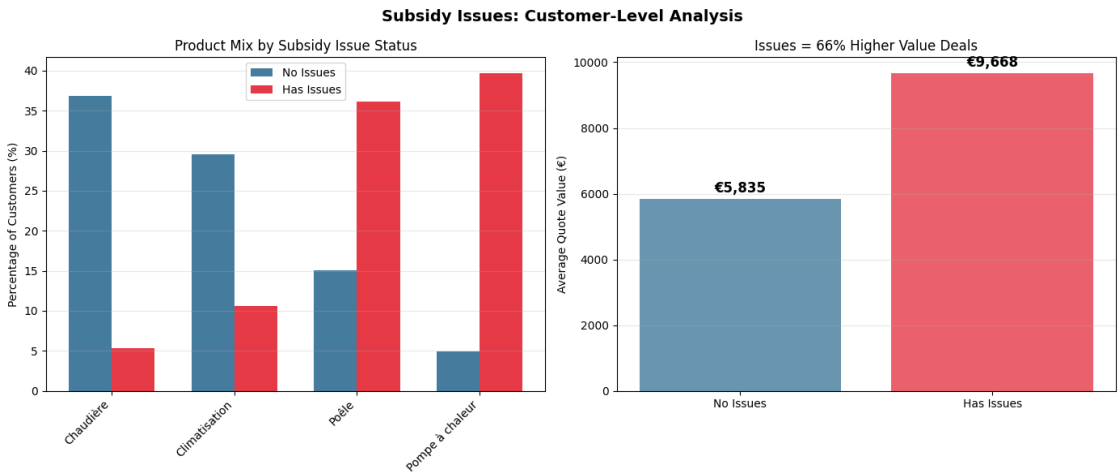

```
Unique values in main_product_family:
main_product_family
Chaudière                          2886
Climatisation                      2475
Poêle                              2033
Pompe à chaleur                    1369
ECS : Chauffe-eau ou adoucisseur    550
Photovoltaïque                      255
Appareil hybride                    109
Emetteur de chauffage  ou chappe     98
Autres                               87
Unknown                              58
Accessoire de pose                   31
Produit VMC                          29
Plomberie Sanitaire                   6
Name: count, dtype: int64

Top products:
main_product_family
Chaudière                          2886
Climatisation                      2475
Poêle                              2033
Pompe à chaleur                    1369
ECS : Chauffe-eau ou adoucisseur    550
Photovoltaïque                      255
Appareil hybride                    109
Emetteur de chauffage  ou chappe     98
Autres                               87
```

```
Unknown                                 58
Name: count, dtype: int64

Using these products: ['Chaudière', 'Climatisation', 'Poêle', 'Pompe à chaleur']

No Issue %: [np.float64(36.856913183279744), np.float64(29.581993569131832),
np.float64(15.032154340836012), np.float64(4.930332261521972)]
Has Issue %: [np.float64(5.35289452815226), np.float64(10.586835844567803),
np.float64(36.12212529738303), np.float64(39.69072164948454)]
```

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================================================================================
Threshold effect
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Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
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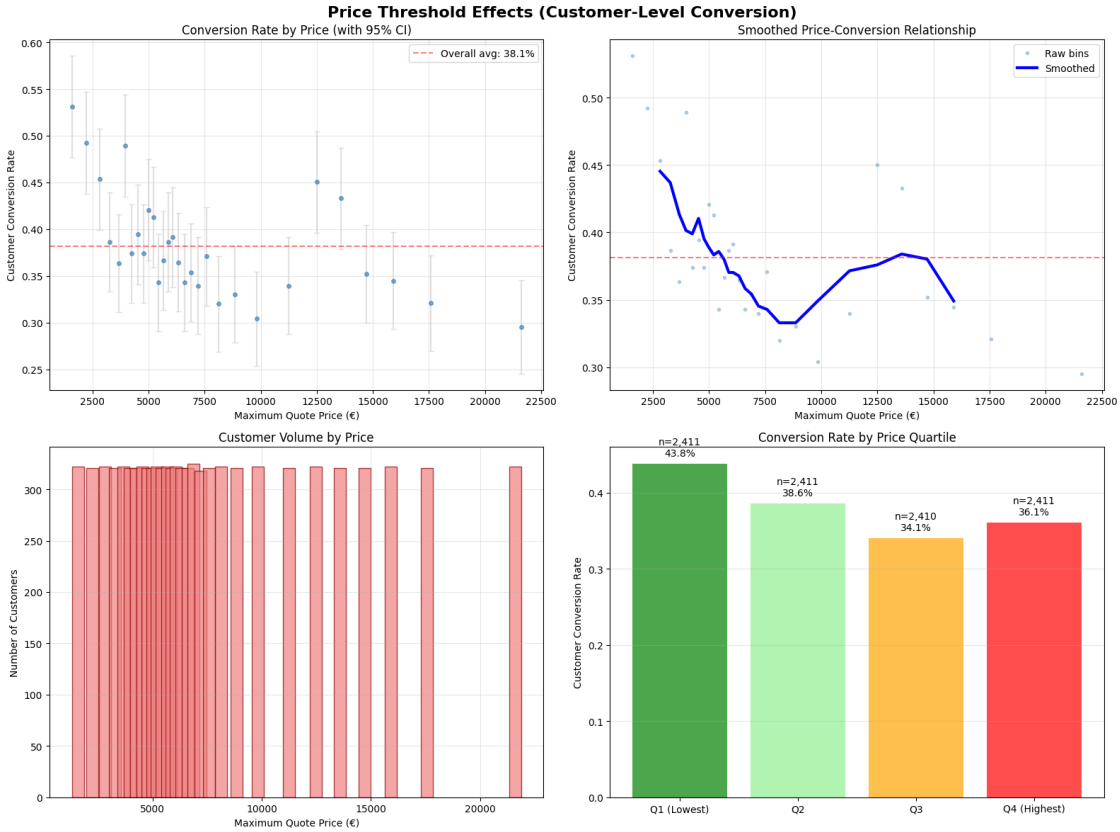


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================================================================================
THRESHOLD EFFECT: KEY FINDINGS
================================================================================

Conversion by Price Quartile:
  Q1 (Lowest): 43.8% (n=2,411)
  Q2: 38.6% (n=2,411)
  Q3: 34.1% (n=2,410)
  Q4 (Highest): 36.1% (n=2,411)

Difference between lowest and highest quartile: 7.7 percentage points

Sweet spots (above-average conversion):
  €1,588: 53.1% (n=322)
  €2,240: 49.2% (n=321)
  €2,816: 45.3% (n=322)
  €3,269: 38.6% (n=321)
  €3,971: 48.9% (n=321)
  €4,530: 39.4% (n=322)
  €4,986: 42.1% (n=321)
```

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  €5,207: 41.3% (n=322)
  €5,880: 38.6% (n=321)
  €6,069: 39.1% (n=322)
  €12,490: 45.0% (n=322)
  €13,582: 43.3% (n=321)

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200

================================================================================
Product mix by price quartile
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Product distribution by price quartile (%):
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
price_quartile
Q1_Low                   31.6    35.1        0.6   25.4    7.3
Q2_MidLow                20.7    36.2        0.5    3.3   39.3
Q3_MidHigh               28.1    34.4        1.7    4.8   31.0
Q4_High                  22.0    11.1       54.2   10.8    2.0

================================================================================
Conversion rates by product and price quartile
================================================================================

Heat Pump (n=1,461):
  Q1_Low: 35.7%
  Q2_MidLow: 46.2%
  Q3_MidHigh: 54.8%
  Q4_High: 40.4%

Boiler (n=2,877):
  Q1_Low: 44.6%
  Q2_MidLow: 39.2%
  Q3_MidHigh: 38.1%
  Q4_High: 31.9%

AC (n=2,522):
  Q1_Low: 41.1%
  Q2_MidLow: 30.5%
  Q3_MidHigh: 29.4%
  Q4_High: 30.4%

Stove (n=2,030):
  Q1_Low: 42.8%
  Q2_MidLow: 41.8%
  Q3_MidHigh: 32.4%
```

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  Q4_High: 36.7%

================================================================================
HEAT PUMP CUSTOMERS (n=1,716)
================================================================================
Average price: €15,977
Conversion rate: 40.7%
Subsidy issue rate: 73.4%

Heat pump adoption by quartile:
  Q1_Low: 0.7%
  Q2_MidLow: 0.6%
  Q3_MidHigh: 2.2%
  Q4_High: 63.9%

================================================================================
Equipment Category Segment Effect
================================================================================

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
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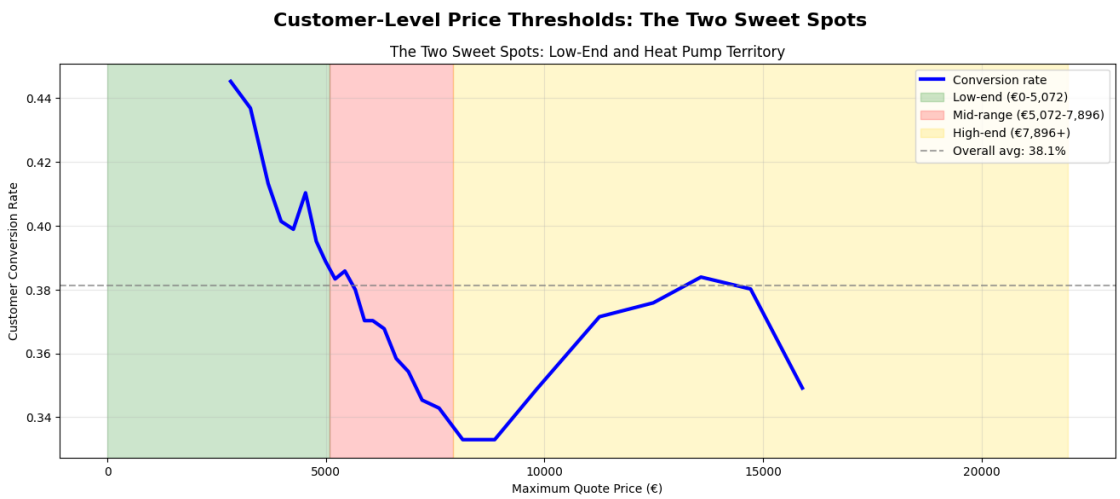

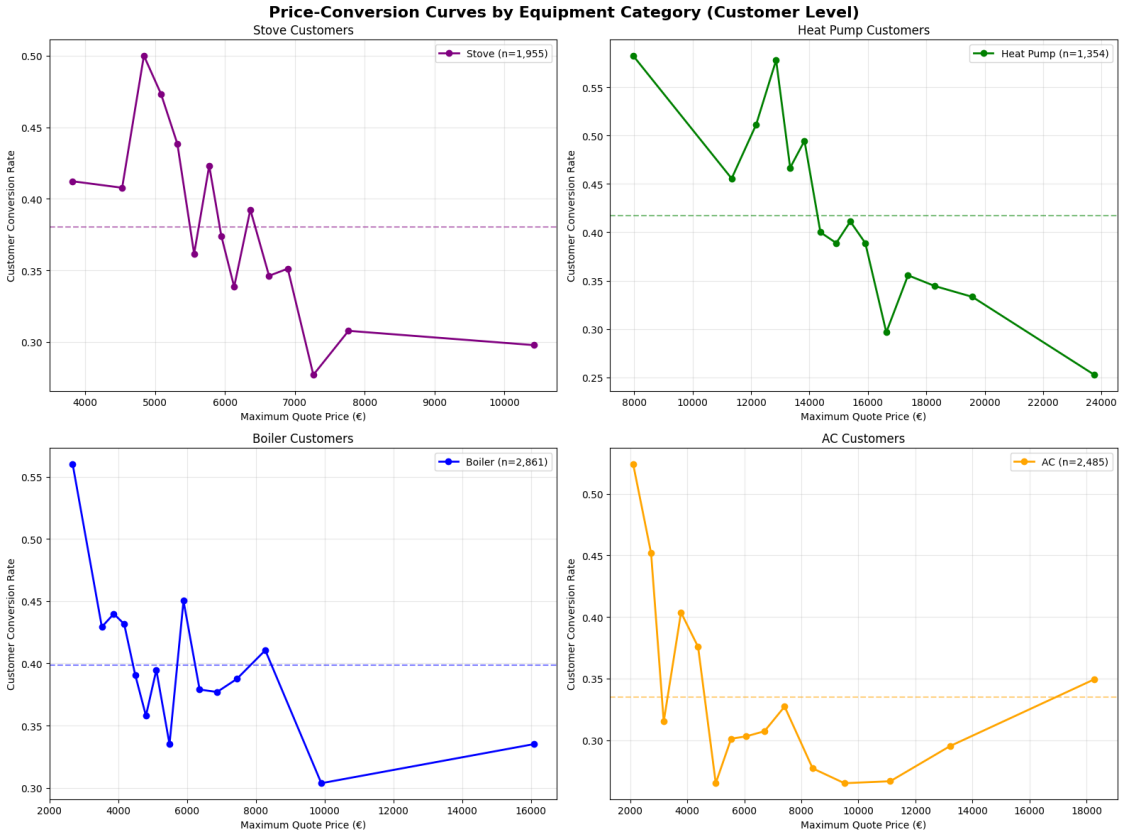


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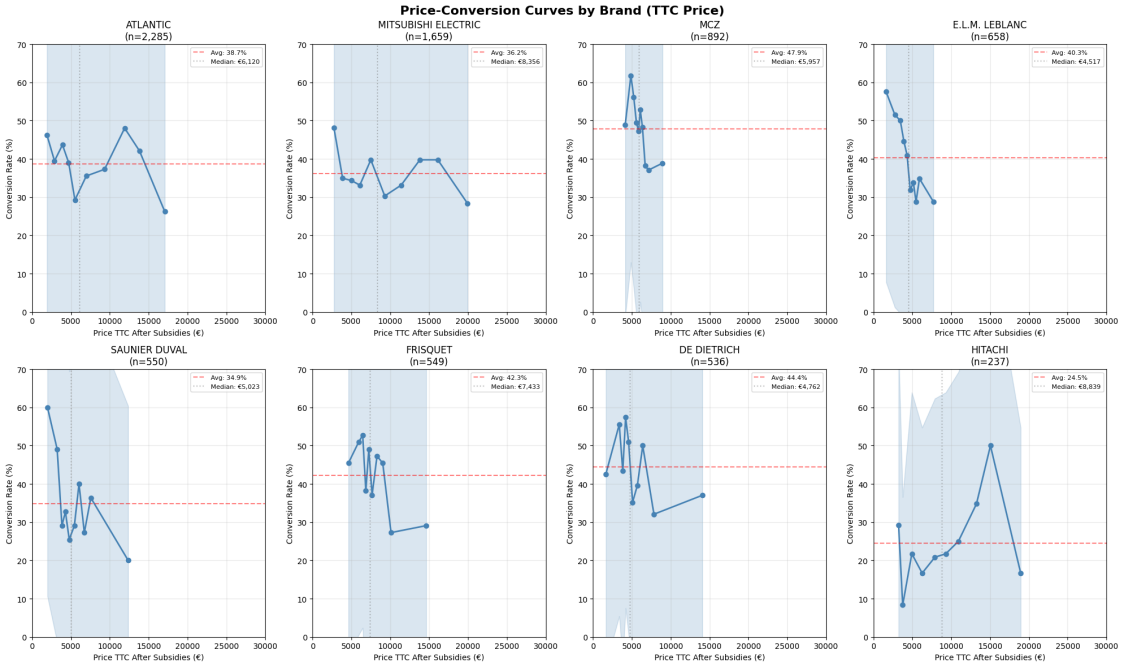
================================================================================
Conversion by Brand Price
================================================================================

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
Top 8 brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'MCZ', 'E.L.M. LEBLANC',
'SAUNIER DUVAL', 'FRISQUET', 'DE DIETRICH', 'HITACHI']
```

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Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
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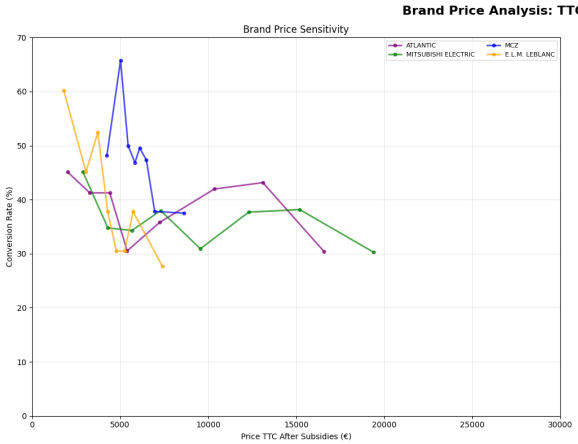

Brand Performance Summary

Brand	Customers	Avg Conv	Avg Price	Best Price Range	Best Conv
ATLANTIC	2,285	38.7%	€7,796	€1,338-3,425	42.9%
MITSUBISHI ELECTRIC	1,659	36.2%	€9,575	€1,368-4,432	41.6%
MCZ	892	47.9%	€6,096	€1,800-5,069	55.3%
E.L.M. LEBLANC	658	40.3%	€4,506	€1,323-3,245	54.5%
SAUNIER DUVAL	550	34.9%	€5,616	€1,332-3,581	54.5%
FRISQUET	549	42.3%	€8,077	€2,391-6,231	48.2%
DE DIETRICH	536	44.4%	€5,676	€3,658-4,428	50.5%
HITACHI	237	24.5%	€9,364	€13,951-25,964	33.3%

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================================================================================
TTC Price-conversion curves: 4 equipment types, multiple brands per graph
================================================================================

Analyzing 9,643 customers
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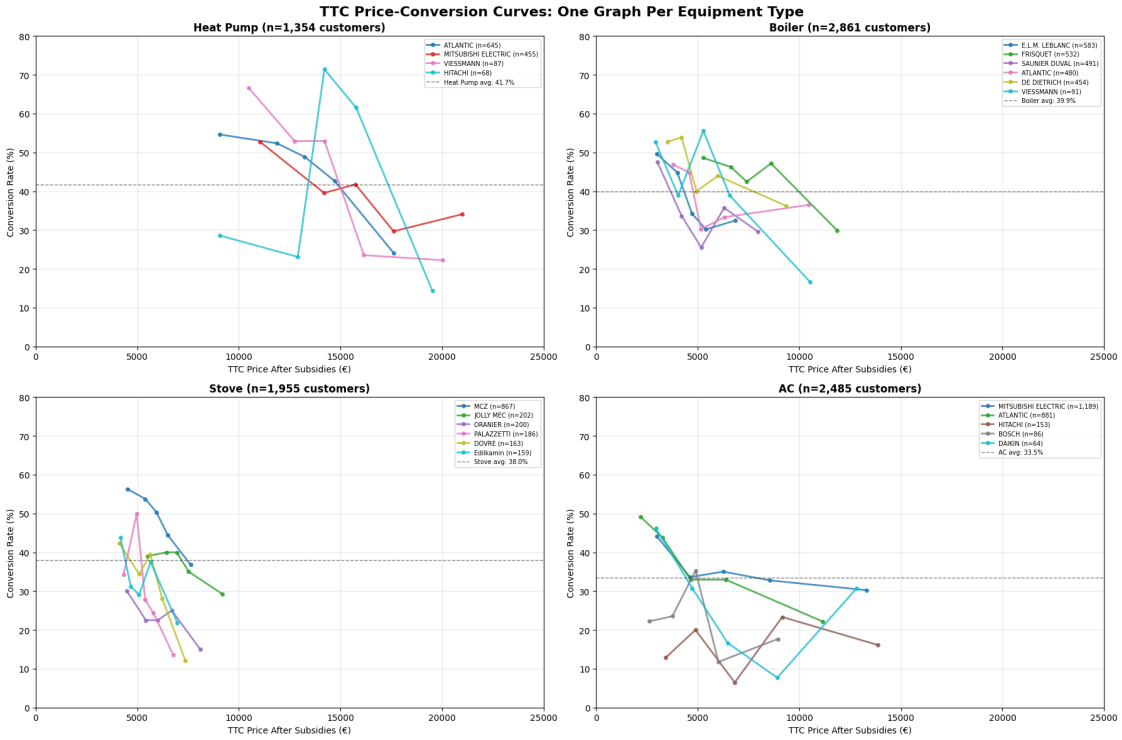
```
Price range: €1,323 to €27,465
Median price: €6,200

Heat Pump - Top brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'VIESSMANN',
'HITACHI']

Boiler - Top brands: ['E.L.M. LEBLANC', 'FRISQUET', 'SAUNIER DUVAL', 'ATLANTIC',
'DE DIETRICH', 'VIESSMANN']

Stove - Top brands: ['MCZ', 'JOLLY MEC', 'ORANIER', 'PALAZZETTI', 'DOVRE',
'Edilkamin']

AC - Top brands: ['MITSUBISHI ELECTRIC', 'ATLANTIC', 'HITACHI', 'BOSCH',
'DAIKIN']
```

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================================================================================
Seasonal Effects on Customer Conversion
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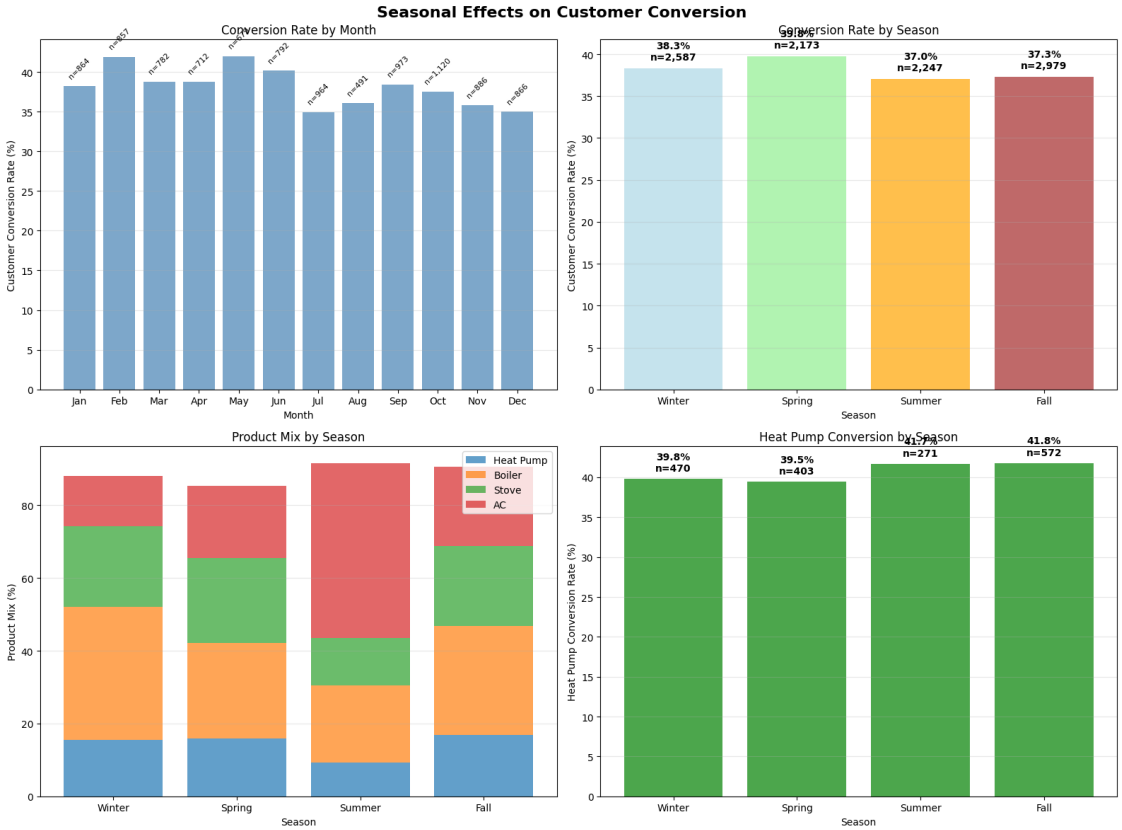


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================================================================================
SEASONAL STATISTICS
================================================================================

Conversion by season:
  Winter: 38.3% (n=2,587)
  Spring: 39.8% (n=2,173)
  Summer: 37.0% (n=2,247)
  Fall: 37.3% (n=2,979)

Seasonality chi-square p-value: 0.2112
Statistically significant: NO

================================================================================
Regional Seasonal Pattern
================================================================================

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
```

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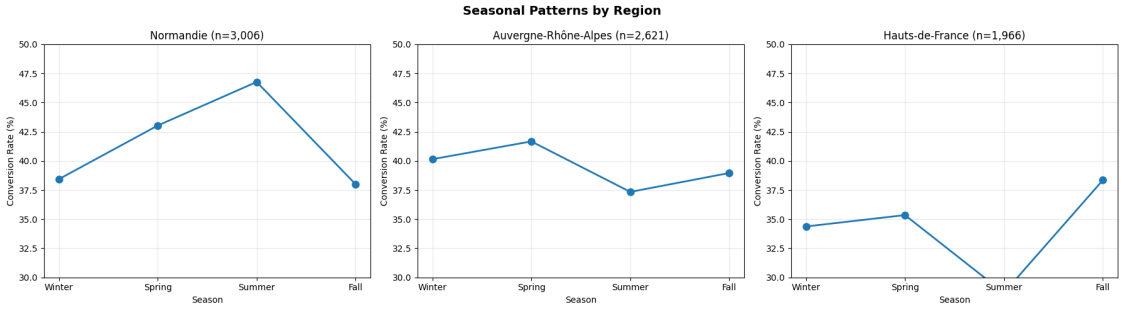
Top regions: ['Normandie', 'Auvergne-Rhône-Alpes', 'Hauts-de-France']
```

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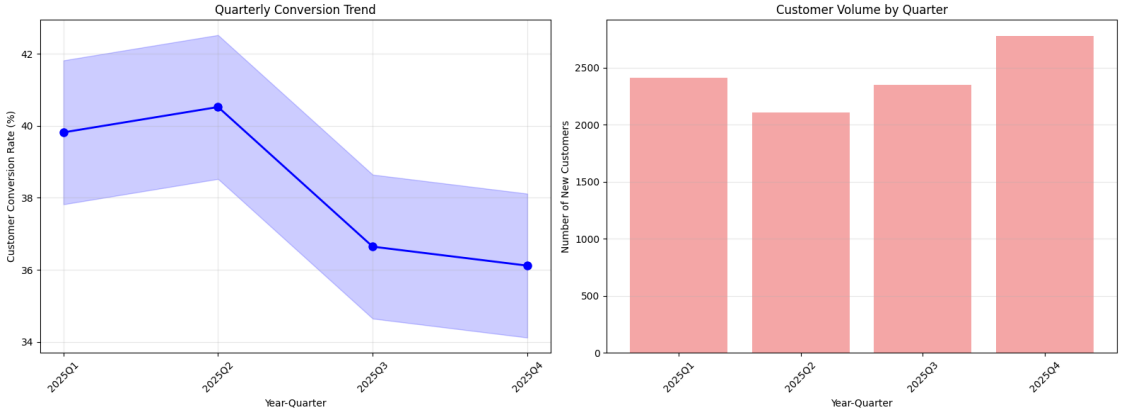
================================================================================
Year-over-year trends
================================================================================

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200

Quarterly conversion rates:
                  mean  count
year_quarter
2025Q1        0.398175   2411
2025Q2        0.405226   2105
2025Q3        0.366425   2347
2025Q4        0.361151   2780
```

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Best and Worst Months Report
================================================================================

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200

Conversion during subsidy suspensions:
                       mean  count
Normal Periods     0.386102   7296
During Suspension  0.366425   2347

Suspension impact p-value: 0.0924
Statistically significant: NO

Best month: 2 (42.3%, n=825)
Worst month: 7 (35.0%, n=934)

Best season: Spring (39.9%)
Worst season: Fall (37.1%)

Heat pump conversion by season:
  Winter: 40.5%
  Spring: 40.4%
  Summer: 43.4%
  Fall: 42.4%

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
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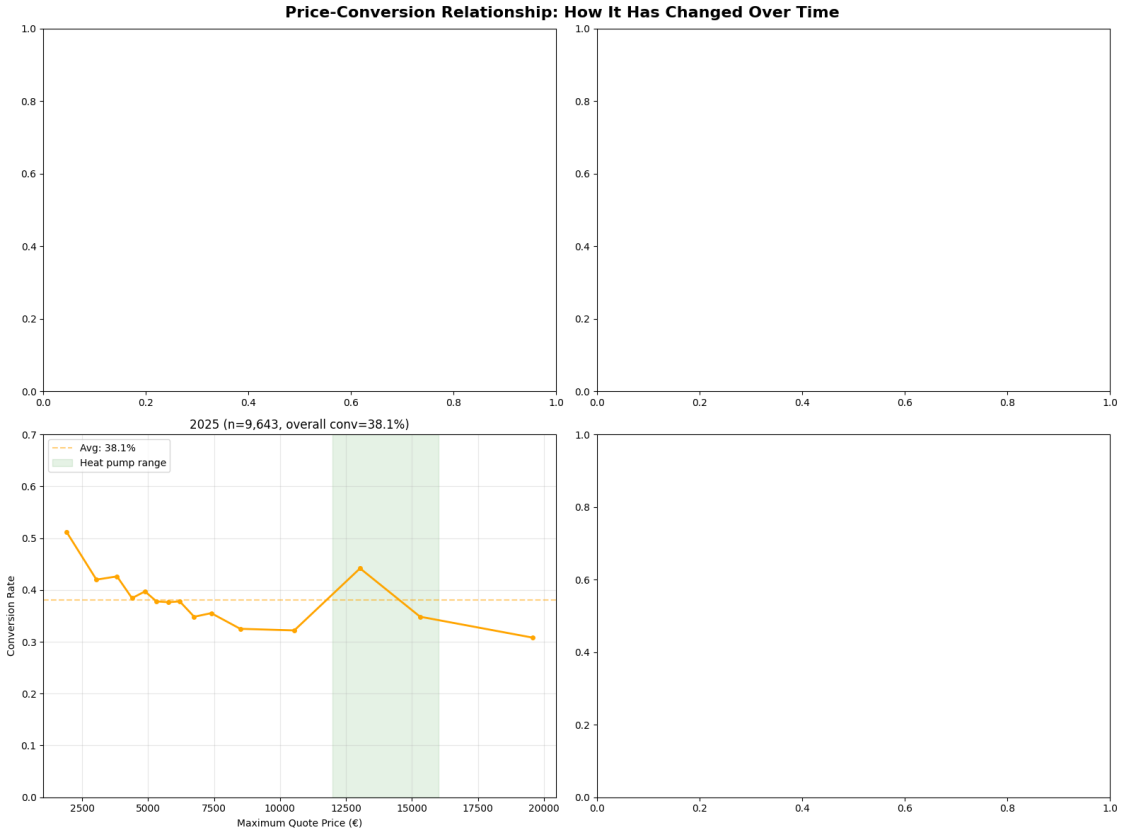


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Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
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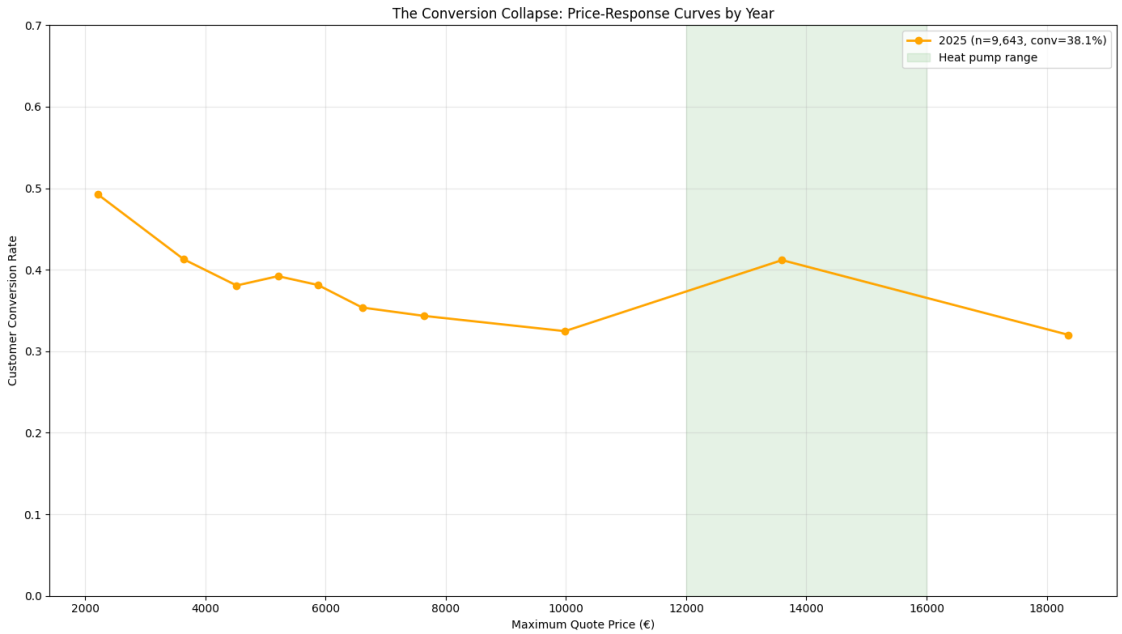


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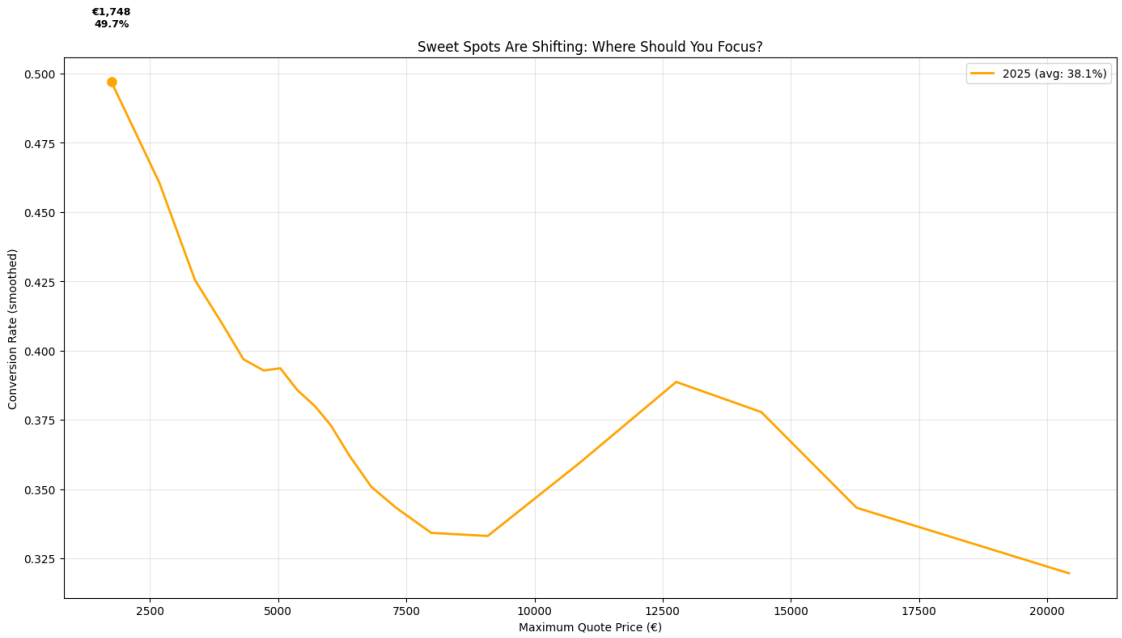
================================================================================
Sweet Spot Analysis Over Time
================================================================================

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200

Sweet spots by year (price points with conversion > avg + 3%):

2025 (overall avg: 38.1%):
  €1,748: 52.4% (n=483)
  €2,682: 46.1% (n=482)
  €3,900: 43.6% (n=482)
  €5,040: 42.1% (n=482)
  €12,766: 44.2% (n=482)
```

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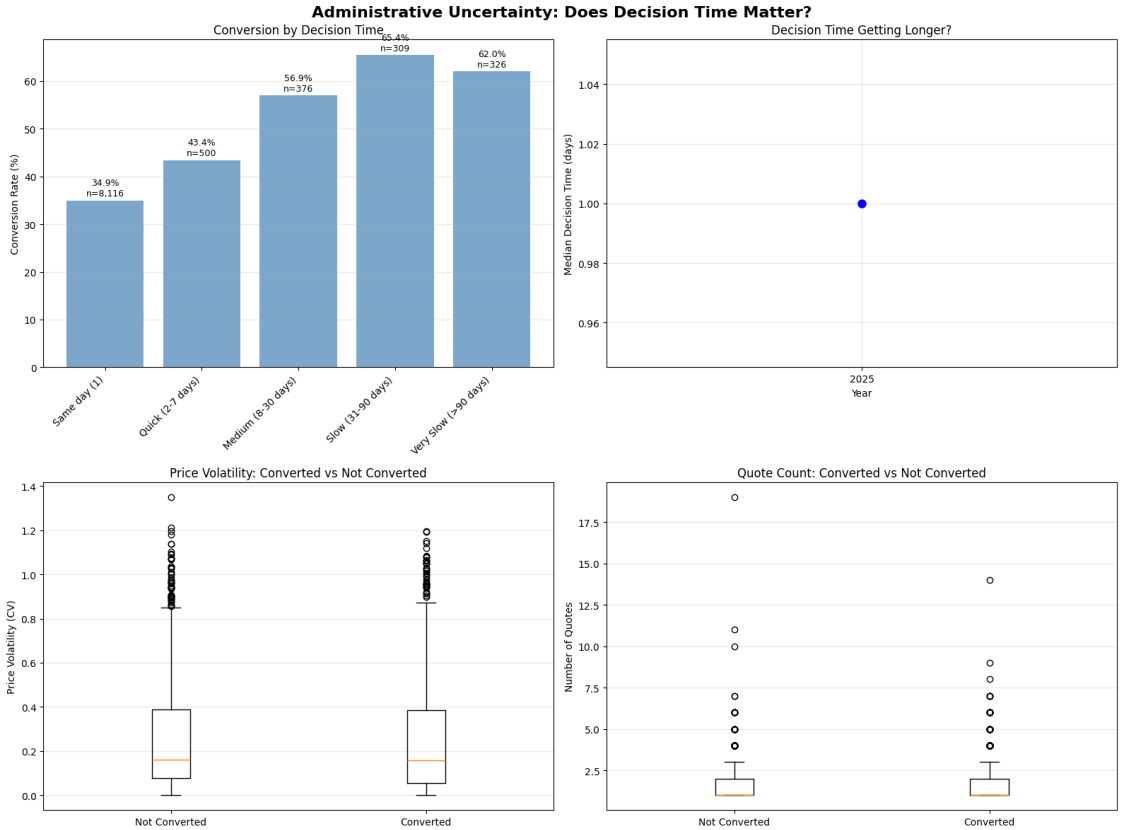
================================================================================
Administrative Uncertainty Effect
================================================================================

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200

Decision time distribution:
count    9643.000000
mean       10.095717
std        39.029109
min         1.000000
25%         1.000000
50%         1.000000
75%         1.000000
max       429.000000
Name: decision_days, dtype: float64

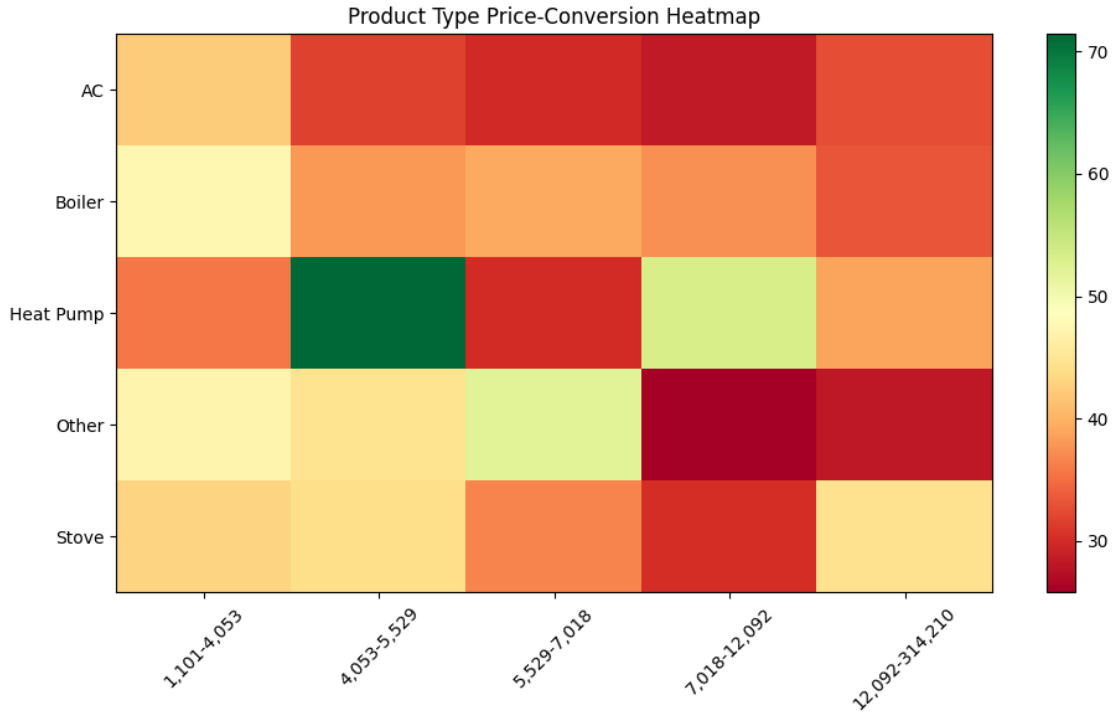
Customers with 1-day decision: 8,116 (84.2%)
```




```
================================================================================
Product Type - Price Conversion Heatmap
================================================================================
Simple pivot table:
price_bin_simple         (1101.459, 4053.15]  (4053.15, 5529.27]  \
main_equipment_category
AC                                 42.336874          31.742243
Boiler                             47.524752          37.922987
Heat Pump                          35.714286          71.428571
Other                              47.019868          44.615385
Stove                              43.023256          44.032258

price_bin_simple         (5529.27, 7018.58]  (7018.58, 12092.49]  \
main_equipment_category
AC                                29.882353           28.378378
Boiler                            39.180328           37.418301
Heat Pump                         30.000000           53.260870
Other                             52.000000           25.781250
Stove                             36.556604           30.158730
```

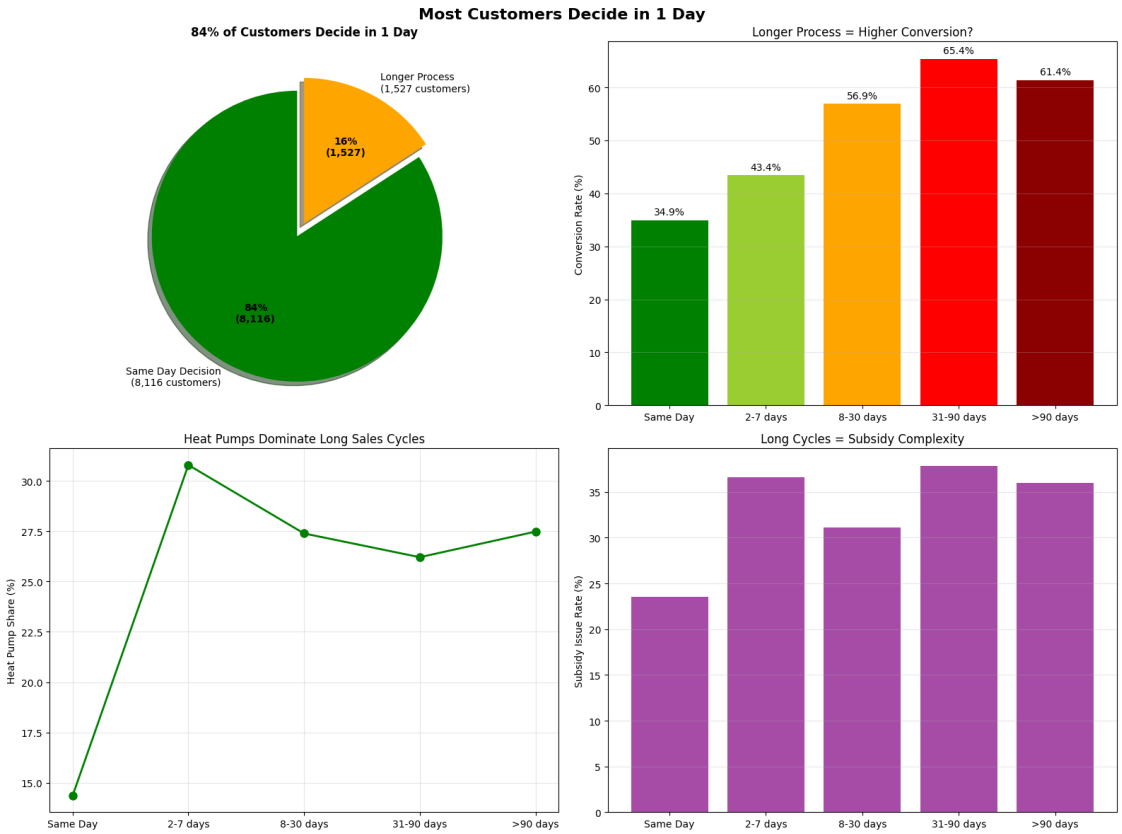
```
price_bin_simple         (12092.49, 314210.61]
main_equipment_category
AC                                   32.564841
Boiler                               33.157895
Heat Pump                            38.837405
Other                                28.110599
Stove                                44.444444
```




```
================================================================================
CHECKING MAIN PRODUCT FAMILY
================================================================================
main_product_family
Chaudière                          2886
Climatisation                      2475
Poêle                              2033
Pompe à chaleur                    1369
ECS : Chauffe-eau ou adoucisseur    550
Photovoltaïque                      255
Appareil hybride                    109
Emetteur de chauffage  ou chappe     98
Autres                               87
Unknown                              58
Accessoire de pose                   31
```

```
Produit VMC                          29
Plomberie Sanitaire                   6
Name: count, dtype: int64

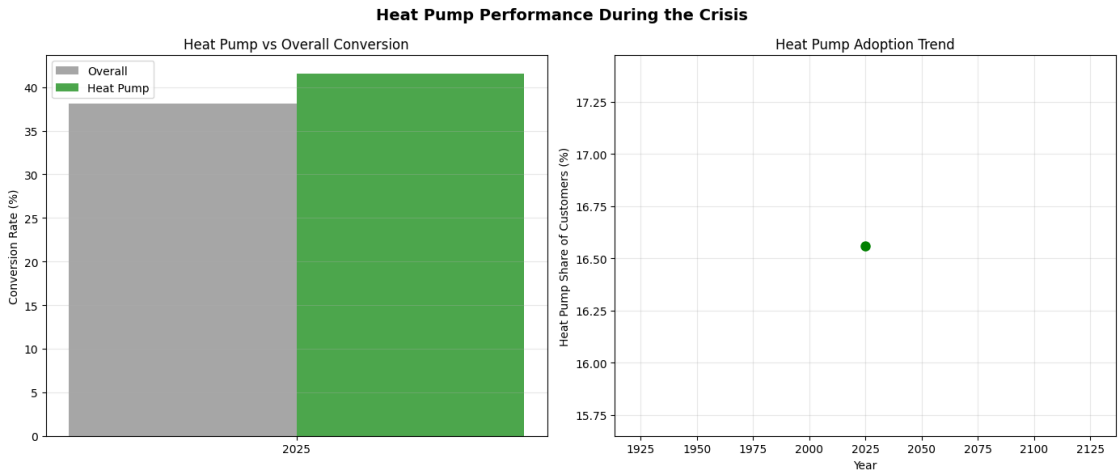
Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200
```




```
================================================================================
Heat Pump Performance Over Time
================================================================================

Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200

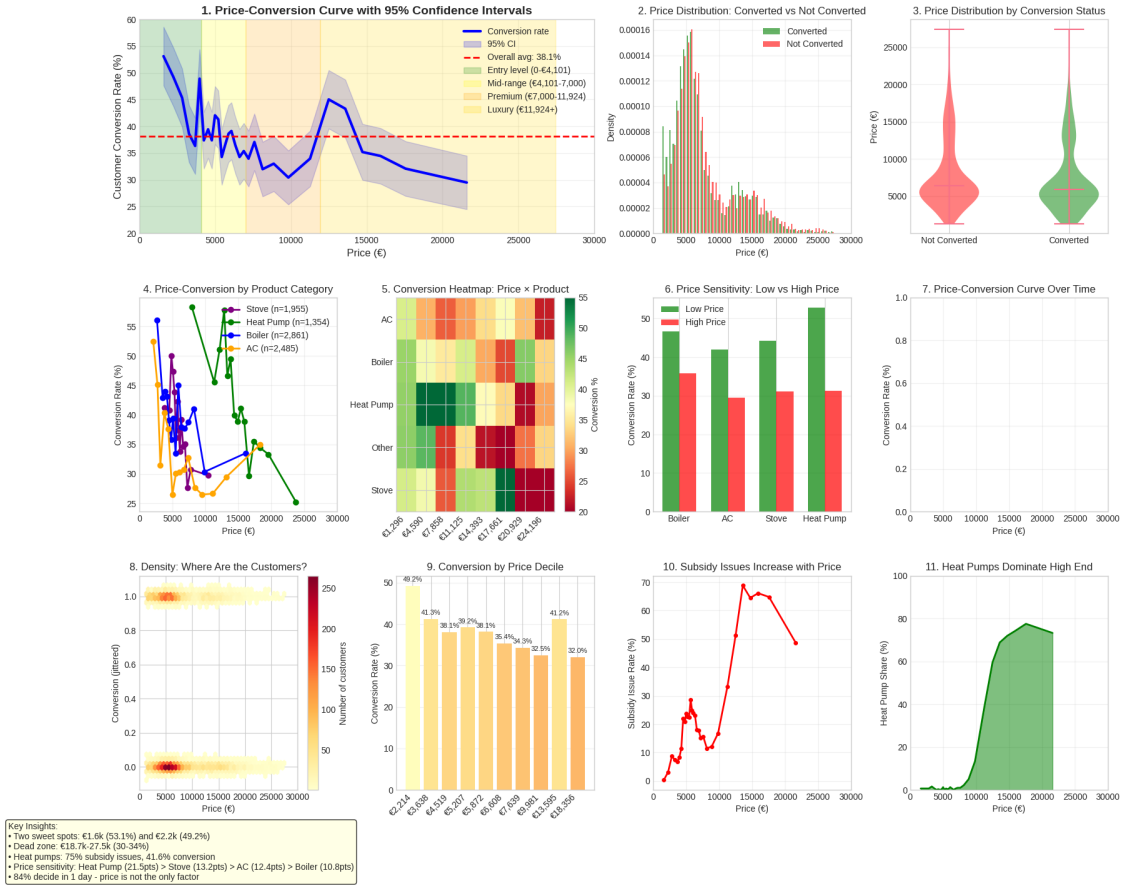
Heat pump customers by year:
        conversion_rate  customer_count  avg_price  subsidy_issue_rate
period
2025              0.416            1597  15227.259               0.753
```



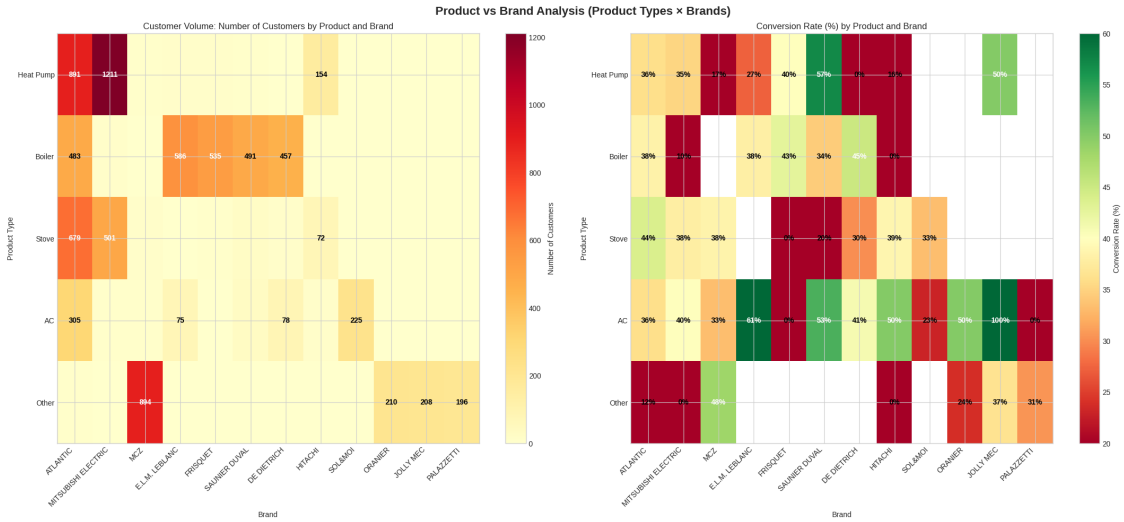

```
Analyzing 9,643 customers
Price range: €1,323 to €27,465
Median price: €6,200

================================================================================
DYNAMIC INSIGHTS CALCULATED:
================================================================================
Sweet spots: €1.6k (53.1%) and €2.2k (49.2%)
Dead zone: €18.7k-27.5k (30-34%)
Heat pumps: 75% subsidy issues, 41.6% conversion
Price sensitivity: Heat Pump (21.5pts) > Stove (13.2pts) > AC (12.4pts) > Boiler
(10.8pts)
Same day decisions: 84%
```

Price-Conversion Dependency Dashboard



Product vs Brand Analysis (Product Types × Brands)



```
================================================================================
PRODUCT × BRAND ANALYSIS
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC  MCZ  E.L.M. LEBLANC
FRISQUET  SAUNIER DUVAL  DE DIETRICH  HITACHI  SOL&MOI  ORANIER  JOLLY MEC
PALAZZETTI
main_equipment_category
AC                            891                 1211    6              11
10              7            1      154        0        0          2           0
Boiler                        483                   10    0             586
535            491          457        1        0        0          0
0
Heat Pump                     679                  501   13               0
8             25           10       72        3        0          0           0
Other                         305                    5    6              75
1             30           78       14      225        2          1           1
Stove                           8                    1  894               0
0              0            0        1        0      210        208         196

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC   MCZ  E.L.M. LEBLANC
FRISQUET  SAUNIER DUVAL  DE DIETRICH  HITACHI  SOL&MOI  ORANIER  JOLLY MEC
PALAZZETTI
main_equipment_category
AC                           36.0                 35.2  16.7            27.3
40.0           57.1          0.0     15.6      NaN      NaN       50.0
NaN
Boiler                       38.3                 10.0   NaN            38.1
42.8           34.4         45.1      0.0      NaN      NaN        NaN
NaN
Heat Pump                    43.9                 37.9  38.5             NaN
0.0            20.0         30.0     38.9     33.3      NaN        NaN
NaN
Other                        36.1                 40.0  33.3            61.3
0.0            53.3         41.0     50.0     23.1     50.0      100.0
0.0
Stove                        12.5                  0.0  48.3             NaN
NaN             NaN          NaN      0.0      NaN     23.8       36.5
30.6

================================================================================
  BEST BRAND BY PRODUCT TYPE
```

```
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: ATLANTIC
  Conversion: 43.9%
  Customers: 679
  Top 3 brands:
    1. ATLANTIC: 43.9% (n=679)
    2. HITACHI: 38.9% (n=72)
    3. MCZ: 38.5% (n=13)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 45.1%
  Customers: 457
  Top 3 brands:
    1. DE DIETRICH: 45.1% (n=457)
    2. FRISQUET: 42.8% (n=535)
    3. ATLANTIC: 38.3% (n=483)

Stove:
  Best brand: MCZ
  Conversion: 48.3%
  Customers: 894
  Top 3 brands:
    1. MCZ: 48.3% (n=894)
    2. JOLLY MEC: 36.5% (n=208)
    3. PALAZZETTI: 30.6% (n=196)

AC:
  Best brand: SAUNIER DUVAL
  Conversion: 57.1%
  Customers: 7
  Top 3 brands:
    1. SAUNIER DUVAL: 57.1% (n=7)
    2. JOLLY MEC: 50.0% (n=2)
    3. FRISQUET: 40.0% (n=10)

Other:
  Best brand: JOLLY MEC
  Conversion: 100.0%
  Customers: 1
  Top 3 brands:
    1. JOLLY MEC: 100.0% (n=1)
    2. E.L.M. LEBLANC: 61.3% (n=75)
    3. SAUNIER DUVAL: 53.3% (n=30)

================================================================================
```

```
  BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Heat Pump
  Conversion: 43.9%
  Customers: 679
  All products:
    Heat Pump: 43.9% (n=679)
    Boiler: 38.3% (n=483)
    Other: 36.1% (n=305)
    AC: 36.0% (n=891)
    Stove: 12.5% (n=8)

MITSUBISHI ELECTRIC:
  Best product: Other
  Conversion: 40.0%
  Customers: 5
  All products:
    Other: 40.0% (n=5)
    Heat Pump: 37.9% (n=501)
    AC: 35.2% (n=1,211)
    Boiler: 10.0% (n=10)
    Stove: 0.0% (n=1)

MCZ:
  Best product: Stove
  Conversion: 48.3%
  Customers: 894
  All products:
    Stove: 48.3% (n=894)
    Heat Pump: 38.5% (n=13)
    Other: 33.3% (n=6)
    AC: 16.7% (n=6)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 61.3%
  Customers: 75
  All products:
    Other: 61.3% (n=75)
    Boiler: 38.1% (n=586)
    AC: 27.3% (n=11)

FRISQUET:
  Best product: Boiler
  Conversion: 42.8%
  Customers: 535
```

```
  All products:
    Boiler: 42.8% (n=535)
    AC: 40.0% (n=10)
    Heat Pump: 0.0% (n=8)
    Other: 0.0% (n=1)

SAUNIER DUVAL:
  Best product: AC
  Conversion: 57.1%
  Customers: 7
  All products:
    AC: 57.1% (n=7)
    Other: 53.3% (n=30)
    Boiler: 34.4% (n=491)
    Heat Pump: 20.0% (n=25)

DE DIETRICH:
  Best product: Boiler
  Conversion: 45.1%
  Customers: 457
  All products:
    Boiler: 45.1% (n=457)
    Other: 41.0% (n=78)
    Heat Pump: 30.0% (n=10)
    AC: 0.0% (n=1)

HITACHI:
  Best product: Other
  Conversion: 50.0%
  Customers: 14
  All products:
    Other: 50.0% (n=14)
    Heat Pump: 38.9% (n=72)
    AC: 15.6% (n=154)
    Boiler: 0.0% (n=1)
    Stove: 0.0% (n=1)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Dominance by Product Category:

Heat Pump:
  Volume leader: ATLANTIC (679 customers, 51.8% share)
  Conversion leader: ATLANTIC (43.9%)

Boiler:
```

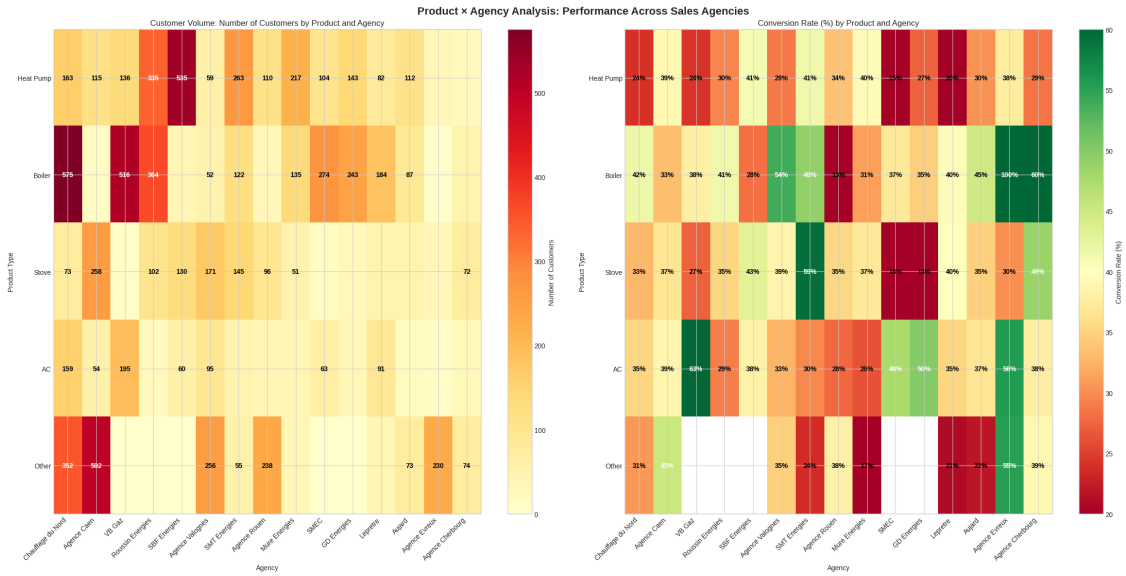
```
  Volume leader: E.L.M. LEBLANC (586 customers, 22.9% share)
  Conversion leader: DE DIETRICH (45.1%)

Stove:
  Volume leader: MCZ (894 customers, 58.9% share)
  Conversion leader: MCZ (48.3%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (1,211 customers, 52.8% share)
  Conversion leader: SAUNIER DUVAL (57.1%)

Other:
  Volume leader: ATLANTIC (305 customers, 41.0% share)
  Conversion leader: JOLLY MEC (100.0%)

Top 15 agencies by customer volume:
  Chauffage du Nord: 1,322 customers, 35.3% conversion
  Agence Caen: 944 customers, 41.6% conversion
  VB Gaz: 858 customers, 41.4% conversion
  Roussin Energies: 825 customers, 35.5% conversion
  SBF Energies: 764 customers, 40.4% conversion
  Agence Valognes: 633 customers, 36.5% conversion
  SMT Energies: 632 customers, 44.6% conversion
  Agence Rouen: 511 customers, 34.4% conversion
  Mure Energies: 477 customers, 34.2% conversion
  SMEC: 458 customers, 33.0% conversion
  GD Energies: 446 customers, 32.1% conversion
  Lepretre: 440 customers, 33.4% conversion
  Aujard: 337 customers, 33.2% conversion
  Agence Evreux: 307 customers, 51.1% conversion
  Agence Cherbourg: 241 customers, 44.8% conversion
```



```
================================================================================
PRODUCT × AGENCY ANALYSIS
================================================================================

  CUSTOMER VOLUME BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              Chauffage du Nord  Agence Caen  VB Gaz  Roussin
Energies  SBF Energies  Agence Valognes  SMT Energies  Agence Rouen  Mure
Energies  SMEC  GD Energies  Lepretre  Aujard  Agence Evreux  Agence Cherbourg
main_equipment_category
AC                                     163          115     136
335           535               59           263           110            217
104          143        82     112             32                21
Boiler                                 575           15     516
364            39               52           122            31            135
274          243       184      87              3                45
Heat Pump                               73          258      11
102           130              171           145            96             51
17           28        45      46             33                72
Other                                  159           54     195
24            60               95           47            36             38
63           32        91      19              9                29
Stove                                  352          502       0
0             0              256           55           238             36
0            0        38      73            230                74

================================================================================
  CONVERSION RATES (%) BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              Chauffage du Nord  Agence Caen  VB Gaz  Roussin
Energies  SBF Energies  Agence Valognes  SMT Energies  Agence Rouen  Mure
Energies  SMEC  GD Energies  Lepretre  Aujard  Agence Evreux  Agence Cherbourg
main_equipment_category
AC                                    23.9         39.1    24.3
29.6          40.9             28.8          41.4          33.6           39.6
15.4         27.3      19.5    30.4           37.5              28.6
Boiler                                41.7         33.3    38.0
41.5          28.2             53.8          49.2          12.9           31.1
37.2         34.6      39.7    44.8          100.0              60.0
Heat Pump                             32.9         36.8    27.3
35.3          43.1             38.6          59.3          35.4           37.3
17.6         14.3      40.0    34.8           30.3              48.6
```

```
Other                                   35.2         38.9    63.1
29.2          38.3             32.6          29.8          27.8           26.3
47.6         50.0      35.2    36.8           55.6              37.9
Stove                                   30.7         45.2     NaN
NaN           NaN             34.8          23.6          38.2           16.7
NaN          NaN      21.1    21.9           55.2              39.2

================================================================================
  BEST AGENCY BY PRODUCT TYPE
--------------------------------------------------------------------------------
--------------------

Heat Pump:
  Best agency: SMT Energies
  Conversion: 59.3%
  Customers: 145
  Top 3 agencies:
    1. SMT Energies: 59.3% (n=145)
    2. Agence Cherbourg: 48.6% (n=72)
    3. SBF Energies: 43.1% (n=130)

Boiler:
  Best agency: Agence Evreux
  Conversion: 100.0%
  Customers: 3
  Top 3 agencies:
    1. Agence Evreux: 100.0% (n=3)
    2. Agence Cherbourg: 60.0% (n=45)
    3. Agence Valognes: 53.8% (n=52)

Stove:
  Best agency: Agence Evreux
  Conversion: 55.2%
  Customers: 230
  Top 3 agencies:
    1. Agence Evreux: 55.2% (n=230)
    2. Agence Caen: 45.2% (n=502)
    3. Agence Cherbourg: 39.2% (n=74)

AC:
  Best agency: SMT Energies
  Conversion: 41.4%
  Customers: 263
  Top 3 agencies:
    1. SMT Energies: 41.4% (n=263)
    2. SBF Energies: 40.9% (n=535)
    3. Mure Energies: 39.6% (n=217)
```

```
Other:
  Best agency: VB Gaz
  Conversion: 63.1%
  Customers: 195
  Top 3 agencies:
    1. VB Gaz: 63.1% (n=195)
    2. Agence Evreux: 55.6% (n=9)
    3. GD Energies: 50.0% (n=32)

================================================================================
 BEST PRODUCT BY AGENCY
--------------------------------------------------------------------------------
--------------------

Chauffage du Nord:
  Best product: Boiler
  Conversion: 41.7%
  Customers: 575
  All products:
    Boiler: 41.7% (n=575)
    Other: 35.2% (n=159)
    Heat Pump: 32.9% (n=73)
    Stove: 30.7% (n=352)
    AC: 23.9% (n=163)

Agence Caen:
  Best product: Stove
  Conversion: 45.2%
  Customers: 502
  All products:
    Stove: 45.2% (n=502)
    AC: 39.1% (n=115)
    Other: 38.9% (n=54)
    Heat Pump: 36.8% (n=258)
    Boiler: 33.3% (n=15)

VB Gaz:
  Best product: Other
  Conversion: 63.1%
  Customers: 195
  All products:
    Other: 63.1% (n=195)
    Boiler: 38.0% (n=516)
    Heat Pump: 27.3% (n=11)
    AC: 24.3% (n=136)

Roussin Energies:
  Best product: Boiler
```

```
  Conversion: 41.5%
  Customers: 364
  All products:
    Boiler: 41.5% (n=364)
    Heat Pump: 35.3% (n=102)
    AC: 29.6% (n=335)
    Other: 29.2% (n=24)

SBF Energies:
  Best product: Heat Pump
  Conversion: 43.1%
  Customers: 130
  All products:
    Heat Pump: 43.1% (n=130)
    AC: 40.9% (n=535)
    Other: 38.3% (n=60)
    Boiler: 28.2% (n=39)

Agence Valognes:
  Best product: Boiler
  Conversion: 53.8%
  Customers: 52
  All products:
    Boiler: 53.8% (n=52)
    Heat Pump: 38.6% (n=171)
    Stove: 34.8% (n=256)
    Other: 32.6% (n=95)
    AC: 28.8% (n=59)

SMT Energies:
  Best product: Heat Pump
  Conversion: 59.3%
  Customers: 145
  All products:
    Heat Pump: 59.3% (n=145)
    Boiler: 49.2% (n=122)
    AC: 41.4% (n=263)
    Other: 29.8% (n=47)
    Stove: 23.6% (n=55)

Agence Rouen:
  Best product: Stove
  Conversion: 38.2%
  Customers: 238
  All products:
    Stove: 38.2% (n=238)
    Heat Pump: 35.4% (n=96)
    AC: 33.6% (n=110)
```

```
    Other: 27.8% (n=36)
    Boiler: 12.9% (n=31)

Mure Energies:
  Best product: AC
  Conversion: 39.6%
  Customers: 217
  All products:
    AC: 39.6% (n=217)
    Heat Pump: 37.3% (n=51)
    Boiler: 31.1% (n=135)
    Other: 26.3% (n=38)
    Stove: 16.7% (n=36)

SMEC:
  Best product: Other
  Conversion: 47.6%
  Customers: 63
  All products:
    Other: 47.6% (n=63)
    Boiler: 37.2% (n=274)
    Heat Pump: 17.6% (n=17)
    AC: 15.4% (n=104)

================================================================================
  AGENCY PERFORMANCE SUMMARY
================================================================================

Top 15 agencies by volume:
                   conversion_rate  customer_count  heat_pump_pct
subsidy_issue_pct  avg_price
main_agency
Chauffage du Nord             35.3            1322            6.8
33.6   6366.251
Agence Caen                   41.6             944           28.9
38.1   8852.018
VB Gaz                        41.4             858            2.2
1.3   4998.778
Roussin Energies              35.5             825           14.4
12.2   7891.331
SBF Energies                  40.4             764           22.3
17.5   9171.634
Agence Valognes               36.5             633           31.9
43.3   9493.014
SMT Energies                  44.6             632           28.6
38.0   6976.193
Agence Rouen                  34.4             511           21.7
43.2   8740.837
```

```
Mure Energies                 34.2            477           13.4
14.0   6855.388
SMEC                          33.0            458            5.0
9.0   6313.018
GD Energies                   32.1            446            9.6
8.7   5723.213
Lepretre                      33.4            440           12.0
25.0   6999.942
Aujard                        33.2            337           15.4
13.6   7706.594
Agence Evreux                 51.1            307           11.7
41.0   6860.088
Agence Cherbourg              44.8            241           32.8
40.2   8655.448

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Agency Excellence by Product Category:

Heat Pump:
  Conversion leader: SMT Energies (59.3%, n=145)

Boiler:
  Conversion leader: Agence Evreux (100.0%, n=3)

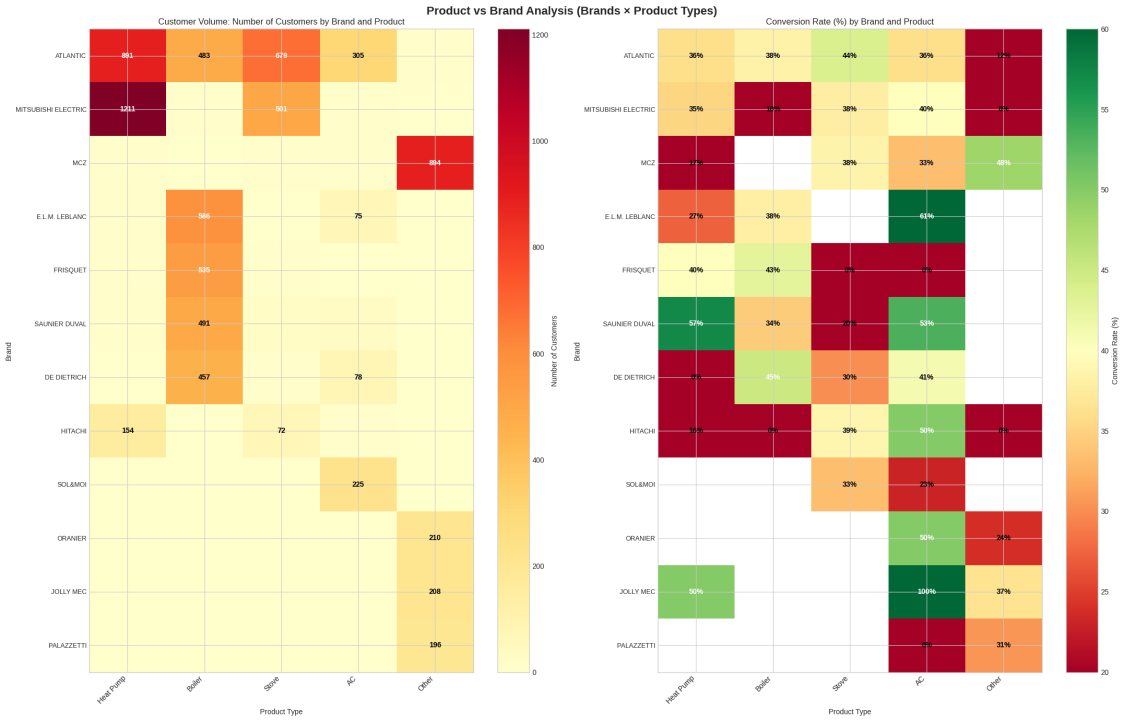
Stove:
  Conversion leader: Agence Evreux (55.2%, n=230)

AC:
  Conversion leader: SMT Energies (41.4%, n=263)

Other:
  Conversion leader: VB Gaz (63.1%, n=195)

================================================================================
 HEAT PUMP SPECIALISTS
================================================================================

Top 5 agencies for Heat Pump conversion:
  SMT Energies: 59.3% (n=145)
  Agence Cherbourg: 48.6% (n=72)
  SBF Energies: 43.1% (n=130)
  Lepretre: 40.0% (n=45)
  Agence Valognes: 38.6% (n=171)
```




```
================================================================================
BRAND × PRODUCT ANALYSIS (Swapped Axes)
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
main_brand
ATLANTIC                  891     483        679    305      8
MITSUBISHI ELECTRIC      1211      10        501      5      1
MCZ                         6       0         13      6    894
E.L.M. LEBLANC             11     586          0     75      0
FRISQUET                   10     535          8      1      0
SAUNIER DUVAL               7     491         25     30      0
DE DIETRICH                 1     457         10     78      0
HITACHI                   154       1         72     14      1
SOL&MOI                     0       0          3    225      0
ORANIER                     0       0          0      2    210
JOLLY MEC                   2       0          0      1    208
PALAZZETTI                  0       0          0      1    196

================================================================================
  CONVERSION RATES (%):
```

```
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
main_brand
ATLANTIC                 36.0    38.3       43.9   36.1   12.5
MITSUBISHI ELECTRIC      35.2    10.0       37.9   40.0    0.0
MCZ                      16.7     NaN       38.5   33.3   48.3
E.L.M. LEBLANC           27.3    38.1        NaN   61.3    NaN
FRISQUET                 40.0    42.8        0.0    0.0    NaN
SAUNIER DUVAL            57.1    34.4       20.0   53.3    NaN
DE DIETRICH               0.0    45.1       30.0   41.0    NaN
HITACHI                  15.6     0.0       38.9   50.0    0.0
SOL&MOI                   NaN     NaN       33.3   23.1    NaN
ORANIER                   NaN     NaN        NaN   50.0   23.8
JOLLY MEC                50.0     NaN        NaN  100.0   36.5
PALAZZETTI                NaN     NaN        NaN    0.0   30.6

================================================================================
  BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Heat Pump
  Conversion: 43.9%
  Customers: 679
  Top 3 products:
    1. Heat Pump: 43.9% (n=679)
    2. Boiler: 38.3% (n=483)
    3. Other: 36.1% (n=305)

MITSUBISHI ELECTRIC:
  Best product: Other
  Conversion: 40.0%
  Customers: 5
  Top 3 products:
    1. Other: 40.0% (n=5)
    2. Heat Pump: 37.9% (n=501)
    3. AC: 35.2% (n=1,211)

MCZ:
  Best product: Stove
  Conversion: 48.3%
  Customers: 894
  Top 3 products:
    1. Stove: 48.3% (n=894)
    2. Heat Pump: 38.5% (n=13)
    3. Other: 33.3% (n=6)

E.L.M. LEBLANC:
```

```
  Best product: Other
  Conversion: 61.3%
  Customers: 75
  Top 3 products:
    1. Other: 61.3% (n=75)
    2. Boiler: 38.1% (n=586)
    3. AC: 27.3% (n=11)

FRISQUET:
  Best product: Boiler
  Conversion: 42.8%
  Customers: 535
  Top 3 products:
    1. Boiler: 42.8% (n=535)
    2. AC: 40.0% (n=10)
    3. Heat Pump: 0.0% (n=8)

SAUNIER DUVAL:
  Best product: AC
  Conversion: 57.1%
  Customers: 7
  Top 3 products:
    1. AC: 57.1% (n=7)
    2. Other: 53.3% (n=30)
    3. Boiler: 34.4% (n=491)

DE DIETRICH:
  Best product: Boiler
  Conversion: 45.1%
  Customers: 457
  Top 3 products:
    1. Boiler: 45.1% (n=457)
    2. Other: 41.0% (n=78)
    3. Heat Pump: 30.0% (n=10)

HITACHI:
  Best product: Other
  Conversion: 50.0%
  Customers: 14
  Top 3 products:
    1. Other: 50.0% (n=14)
    2. Heat Pump: 38.9% (n=72)
    3. AC: 15.6% (n=154)

SOL&MOI:
  Best product: Heat Pump
  Conversion: 33.3%
  Customers: 3
```

```
  Top 3 products:
    1. Heat Pump: 33.3% (n=3)
    2. Other: 23.1% (n=225)

ORANIER:
  Best product: Other
  Conversion: 50.0%
  Customers: 2
  Top 3 products:
    1. Other: 50.0% (n=2)
    2. Stove: 23.8% (n=210)

================================================================================
 BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: ATLANTIC
  Conversion: 43.9%
  Customers: 679
  Top 5 brands:
    1. ATLANTIC: 43.9% (n=679)
    2. HITACHI: 38.9% (n=72)
    3. MCZ: 38.5% (n=13)
    4. MITSUBISHI ELECTRIC: 37.9% (n=501)
    5. SOL&MOI: 33.3% (n=3)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 45.1%
  Customers: 457
  Top 5 brands:
    1. DE DIETRICH: 45.1% (n=457)
    2. FRISQUET: 42.8% (n=535)
    3. ATLANTIC: 38.3% (n=483)
    4. E.L.M. LEBLANC: 38.1% (n=586)
    5. SAUNIER DUVAL: 34.4% (n=491)

Stove:
  Best brand: MCZ
  Conversion: 48.3%
  Customers: 894
  Top 5 brands:
    1. MCZ: 48.3% (n=894)
    2. JOLLY MEC: 36.5% (n=208)
    3. PALAZZETTI: 30.6% (n=196)
    4. ORANIER: 23.8% (n=210)
    5. ATLANTIC: 12.5% (n=8)
```

```
AC:
  Best brand: SAUNIER DUVAL
  Conversion: 57.1%
  Customers: 7
  Top 5 brands:
    1. SAUNIER DUVAL: 57.1% (n=7)
    2. JOLLY MEC: 50.0% (n=2)
    3. FRISQUET: 40.0% (n=10)
    4. ATLANTIC: 36.0% (n=891)
    5. MITSUBISHI ELECTRIC: 35.2% (n=1,211)

Other:
  Best brand: JOLLY MEC
  Conversion: 100.0%
  Customers: 1
  Top 5 brands:
    1. JOLLY MEC: 100.0% (n=1)
    2. E.L.M. LEBLANC: 61.3% (n=75)
    3. SAUNIER DUVAL: 53.3% (n=30)
    4. ORANIER: 50.0% (n=2)
    5. HITACHI: 50.0% (n=14)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Performance by Product Category:

ATLANTIC:
  Heat Pump: 43.9% (n=679)
  Boiler: 38.3% (n=483)
  Other: 36.1% (n=305)

MITSUBISHI ELECTRIC:
  Other: 40.0% (n=5)
  Heat Pump: 37.9% (n=501)
  AC: 35.2% (n=1,211)

MCZ:
  Stove: 48.3% (n=894)
  Heat Pump: 38.5% (n=13)
  Other: 33.3% (n=6)

E.L.M. LEBLANC:
  Other: 61.3% (n=75)
  Boiler: 38.1% (n=586)
  AC: 27.3% (n=11)
```

```
FRISQUET:
  Boiler: 42.8% (n=535)
  AC: 40.0% (n=10)
  Heat Pump: 0.0% (n=8)

SAUNIER DUVAL:
  AC: 57.1% (n=7)
  Other: 53.3% (n=30)
  Boiler: 34.4% (n=491)

DE DIETRICH:
  Boiler: 45.1% (n=457)
  Other: 41.0% (n=78)
  Heat Pump: 30.0% (n=10)

HITACHI:
  Other: 50.0% (n=14)
  Heat Pump: 38.9% (n=72)
  AC: 15.6% (n=154)

================================================================================
 MARKET LEADERS BY PRODUCT CATEGORY
================================================================================

Heat Pump:
  Volume leader: ATLANTIC (679 customers, 51.8% share)
  Conversion leader: ATLANTIC (43.9%)

Boiler:
  Volume leader: E.L.M. LEBLANC (586 customers, 22.9% share)
  Conversion leader: DE DIETRICH (45.1%)

Stove:
  Volume leader: MCZ (894 customers, 58.9% share)
  Conversion leader: MCZ (48.3%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (1,211 customers, 52.8% share)
  Conversion leader: SAUNIER DUVAL (57.1%)

Other:
  Volume leader: ATLANTIC (305 customers, 41.0% share)
  Conversion leader: JOLLY MEC (100.0%)

 (main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
  main_brand
```

```
ATLANTIC                  891     483        679    305      8
MITSUBISHI ELECTRIC      1211      10        501      5      1
MCZ                         6       0         13      6    894
E.L.M. LEBLANC             11     586          0     75      0
FRISQUET                   10     535          8      1      0
SAUNIER DUVAL               7     491         25     30      0
DE DIETRICH                 1     457         10     78      0
HITACHI                   154       1         72     14      1
SOL&MOI                     0       0          3    225      0
ORANIER                     0       0          0      2    210
JOLLY MEC                   2       0          0      1    208
PALAZZETTI                  0       0          0      1    196,
main_equipment_category         AC     Boiler  Heat Pump       Other  \
main_brand
ATLANTIC                 36.026936  38.302277  43.888071   36.065574
MITSUBISHI ELECTRIC      35.177539  10.000000  37.924152   40.000000
MCZ                      16.666667        NaN  38.461538   33.333333
E.L.M. LEBLANC           27.272727  38.054608        NaN   61.333333
FRISQUET                 40.000000  42.803738   0.000000    0.000000
SAUNIER DUVAL            57.142857  34.419552  20.000000   53.333333
DE DIETRICH               0.000000  45.076586  30.000000   41.025641
HITACHI                  15.584416   0.000000  38.888889   50.000000
SOL&MOI                        NaN        NaN  33.333333   23.111111
ORANIER                        NaN        NaN        NaN   50.000000
JOLLY MEC                50.000000        NaN        NaN  100.000000
PALAZZETTI                     NaN        NaN        NaN    0.000000

main_equipment_category      Stove
main_brand
ATLANTIC                 12.500000
MITSUBISHI ELECTRIC       0.000000
MCZ                      48.322148
E.L.M. LEBLANC                 NaN
FRISQUET                       NaN
SAUNIER DUVAL                  NaN
DE DIETRICH                    NaN
HITACHI                   0.000000
SOL&MOI                        NaN
ORANIER                  23.809524
JOLLY MEC                36.538462
PALAZZETTI               30.612245  )
```

```python
import ipynbname
import subprocess
import os
import sys
```

```python
def export_notebook_to_pdf(overwrite=True, verbose=True):
    """
    Export current notebook to PDF with overwrite control

    Args:
        overwrite (bool): If True, overwrite existing PDF file
        verbose (bool): If True, print progress messages

    Returns:
        bool: True if successful, False otherwise
    """
    try:
        # Get notebook path
        notebook_path = ipynbname.path()
        notebook_path_str = str(notebook_path)
        pdf_path = notebook_path_str.replace('.ipynb', '.pdf')

        if verbose:
            print(f"  Notebook: {notebook_path.name}")
            print(f"  PDF path: {pdf_path}")

        # Check if PDF exists
        if os.path.exists(pdf_path):
            if overwrite:
                if verbose:
                    print(f"  Removing existing PDF: {pdf_path}")
                os.remove(pdf_path)
            else:
                if verbose:
                    print(f"  PDF already exists and overwrite=False")
                return False

        # Convert to PDF
        if verbose:
            print("  Converting to PDF...")

        result = subprocess.run([
            'jupyter', 'nbconvert', '--to', 'pdf',
            '--no-prompt',  # Suppress input prompts
            notebook_path_str
        ], capture_output=True, text=True)

        if result.returncode == 0:
            if verbose:
                print(f"  PDF saved: {pdf_path}")
            return True
        else:
```

```
            if verbose:
                print(f"  Conversion failed:")
                print(result.stderr)
            return False

    except Exception as e:
        if verbose:
            print(f"  Error: {e}")
        return False

# Use the function
if __name__ == "__main__":
    # Overwrite existing PDF
    export_notebook_to_pdf(overwrite=True)

    # Or create new version without overwriting
    # export_notebook_to_pdf(overwrite=False)export_full_notebook_to_pdf()
```

```
 Notebook: _eda_price_2025.ipynb
 PDF path: /home/valeriya/project/homeserve/ai-france-hvac/_eda_price_2025.pdf
 Removing existing PDF: /home/valeriya/project/homeserve/ai-france-
hvac/_eda_price_2025.pdf
 Converting to PDF…
 PDF saved: /home/valeriya/project/homeserve/ai-france-hvac/_eda_price_2025.pdf
```